



KEYSTRA ENERGY GROUP ENTERPRISE

SOLUTION 2: PIPELINE INTEGRITY AND OPERATIONAL RISK

Pipeline Incident Analytics and Integrity Intelligence

BUSINESS QUESTION:

How can historical pipeline incident data can be used to identify operational risk, understand why incidents occur, and prioritise pipeline assets for integrity management?

PROJECT OBJECTIVE:

To develop a solution that transforms historical pipeline incident data into actionable operational intelligence. The solution helps pipeline operators identify high risk assets, understand the causes and consequences of incidents, and support proactive integrity management and maintenance decisions.

KEY ANALYTICS AREA:

- Pipeline Incident Trend Analysis
- Incident Cause Analysis
- Asset Integrity Assessment
- Geographic Risk Assessment
- Operational and Financial Impact Analysis
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In Solution 1 — Enterprise HSE Performance Management, the focus was on strengthening organisational safety performance by using data to monitor incidents, exposures, corrective actions, and emerging HSE risks across Keystra Energy Group. While enterprise HSE performance management plays an important role in protecting people and improving safety outcomes, reliable energy operations also depend on the integrity of the infrastructure that transports energy products safely from one location to another. This makes pipeline integrity another critical component of Keystra Energy Group's enterprise asset management strategy.

Unlike equipment that is continuously monitored through operational sensors, pipeline systems are often assessed through inspections and historical incident records. Every reported incident provides valuable information about the condition of the pipeline network, the assets involved, the circumstances surrounding the failure, and the consequences that followed. When examined collectively, these incidents reveal patterns that help organisations understand where operational risk exists and where integrity efforts should be focused.

To strengthen this aspect of asset management, Keystra Energy Group developed Solution 2— Pipeline Integrity & Incident Risk Intelligence using the PHMSA Hazardous Liquid Accident dataset. The solution investigates historical pipeline incidents to understand how risk is distributed across the network, identify the factors contributing to failures, evaluate the impact of those failures, and support more informed integrity management decisions. Rather than functioning as a compliance reporting exercise, this solution demonstrates how historical incident data can be transformed into a strategic business asset.

By providing a clearer understanding of pipeline risk, it enables integrity engineers, operations managers, and executive leaders to prioritise inspections, direct maintenance resources more effectively, and make better informed decisions that improve the long-term reliability and safety of pipeline operations.

BUSINESS CONTEXT

Pipeline operators manage extensive networks that vary in age, design, operating conditions, and environmental exposure. Maintaining the integrity of these assets requires continuous decisions about inspections, maintenance, repairs, and capital investment. Because resources are limited, these decisions cannot be based on assumptions alone, they must be guided by a clear understanding of where risk is greatest.

Historical pipeline incidents provide an important source of that understanding. Beyond documenting what happened, they help explain where incidents occur most frequently, which assets are repeatedly involved, the failure mechanisms responsible, and the operational, environmental, and safety consequences that follow. Analysing these patterns allows organisations to identify recurring risks, prioritise integrity activities, and focus preventive efforts where they are likely to deliver the greatest value.

Recognising this need, Keystra Energy Group initiated this solution to answer a fundamental business question: How can historical pipeline incident data be used to support better integrity management decisions? Addressing this question forms the foundation of the analysis presented throughout this solution.

BUSINESS OBJECTIVES

The objective of this solution is to develop an enterprise analytics solution that helps pipeline operators:

- * Understand where operational risk exists across the pipeline network.
- * Identify the factors contributing to pipeline incidents.
- * Evaluate the financial, environmental, and safety consequences of those incidents.
- * Determine which assets and operating conditions require greater integrity management attention.
- * Support inspection planning, maintenance prioritisation, and preventive investment using historical operational evidence.

BUSINESS VALUE

Pipeline incidents can have consequences that extend well beyond the point of failure. In addition to repair costs, they can disrupt operations, result in product loss, impact the environment, increase regulatory scrutiny, and pose risks to public and employee safety. These outcomes make pipeline integrity more than an engineering responsibility, it is a business priority that directly influences operational resilience, financial performance, and organisational reputation.

This solution enables Keystra Energy Group to move beyond simply recording incidents by using historical operational data to better understand where risks exist, what factors

contribute to failures, and which assets require greater attention. By providing this broader view of pipeline risk, the organisation can make more informed decisions about inspection planning, maintenance prioritisation, and preventive investment.

Ultimately, the value of the solution lies in its ability to support a more proactive approach to pipeline integrity management. Rather than responding to failures after they occur, decision makers are equipped with the information needed to anticipate areas of concern, focus resources where they will have the greatest impact, and strengthen the long-term safety, reliability, and performance of the pipeline network.

The Business Understanding phase established what this solution was expected to achieve. The next step was to determine whether the available data could provide the level of detail needed to answer those business questions. Before any analysis could begin, it was important to understand what information was available, how it had been collected, and whether it was suitable for investigating pipeline integrity and operational risk.

DATASET OVERVIEW

To support this analysis, Keystra Energy Group **selected** the PHMSA Hazardous Liquid Accident Data (January 2010 – Present), a real-world regulatory dataset published by the Pipeline and Hazardous Materials Safety Administration (PHMSA), an agency of the U.S. Department of Transportation. The dataset contains reported hazardous liquid pipeline incidents submitted by operators across the United States in accordance with federal reporting requirements. Because it is continuously updated as new incidents are reported and investigated, it provides an extensive and current record of pipeline accident history.

The version used in this project contains incidents reported between January 2010 and 2026, providing more than fifteen years of historical operational data. This broad reporting period was particularly important for the objectives of Solution 3.

Rather than examining individual incidents in isolation, the project seeks to understand how pipeline incidents have changed over time, where operational risk is concentrated, which assets experience the greatest exposure, what causes failures, and the consequences those failures create. Answering these questions requires a dataset with sufficient historical depth, and the PHMSA dataset provides that foundation.

Another reason for selecting this dataset is the richness of the information captured for each reported incident. Every incident record goes beyond documenting that an accident occurred. It describes when the incident happened, where it occurred, which operator was involved, the type of pipeline facility, the affected system component, the reported cause, pipeline characteristics such as diameter and installation year, as well as the financial, environmental, and safety outcomes of the event. This level of detail makes it possible to investigate pipeline incidents from multiple business perspectives rather than focusing on a single measure of performance.

The dataset also aligns closely with the objectives established during the Business Understanding phase. It contains the information needed to analyse incident trends, evaluate geographic and operator exposure, assess asset integrity, investigate failure mechanisms, and understand the operational consequences of pipeline incidents. These capabilities made it an appropriate foundation for developing an enterprise pipeline integrity and incident risk intelligence solution.

While the dataset provides a comprehensive view of reported pipeline incidents, it was originally designed for regulatory reporting rather than business analytics. As a result, the raw data contains a large number of fields that are not directly relevant to this investigation. Before meaningful analysis could begin, the dataset needed to be carefully reviewed, refined, and prepared to ensure it aligned with the business objectives of the project. The next section describes this preparation process.

DATA PREPARATION

Once the initial review was complete, attention shifted to preparing the data for the next stage of the project. Although the PHMSA dataset offered extensive information on hazardous liquid pipeline incidents, it had been designed for regulatory reporting rather than analytical use. This meant that, before any SQL work could begin, the data needed to be organised into a structure that reflected the needs of this solution. Preparation was carried out in Microsoft Excel using Power Query. The first step was to profile the entire dataset, providing a complete view of its structure, quality, and completeness. This confirmed that the source file contained 5,959 reported incidents across 648 variables, capturing a broad range of operational, engineering, environmental, financial, and administrative information.

With a clear understanding of the data, each variable was assessed to determine its practical value to the project. Priority was given to information describing incident timing, geographic location, pipeline operators, facility types, system components, pipeline characteristics, failure causes, product release, environmental impact, financial losses, and safety outcomes. As part of this review, data types were validated, missing values were examined, and duplicate incident identifiers were checked to improve the reliability of the working data.

The review also identified a significant number of fields that existed solely to support regulatory documentation and administrative reporting. Since these attributes would not contribute to the analytical questions being explored, they were excluded from the working model. This allowed the project to concentrate on the information most relevant to understanding incident trends, operational risk, asset performance, and incident consequences. Following this refinement, the number of variables was reduced from 648 to 38, resulting in a streamlined analytical dataset that preserved the information required for the remaining phases of the project.

The completed dataset was then exported to MySQL, where it served as the foundation for database engineering, SQL based analysis, and executive dashboard development.

DATABASE ENGINEERING

The data preparation phase marked an important milestone in the development of this solution. The project now had a focused analytical dataset containing the information

required to investigate pipeline integrity and operational risk. However, before meaningful analysis could begin, Keystra Energy Group needed to establish a reliable analytical environment capable of supporting structured SQL investigation, data validation, and executive reporting. Rather than performing the analysis directly in Excel, the prepared dataset was exported into MySQL, where a dedicated analytical database was developed. This provided a controlled environment for managing the data, executing complex business queries, and maintaining a clear separation between the prepared source data and the analytical processes built on top of it. Creating the database also reflected a common enterprise analytics practice, where data preparation, storage, analysis, and reporting are managed through distinct but connected stages.

Following the import, the database underwent a series of validation checks to confirm that the data had been transferred accurately and was suitable for analysis. Record counts were verified, field structures were reviewed, and key data types were validated to ensure consistency across the analytical environment. During this process, a source formatting issue affecting the CAUSE field was identified and resolved before any SQL investigation commenced. Addressing this issue at an early stage helped preserve the reliability of subsequent queries and prevented inconsistencies from carrying through to the reporting layer. The prepared dataset initially contained 5,959 incident records. During database validation, 17 records (0.29%) containing source formatting inconsistencies could not be successfully parsed and were therefore excluded from the analytical database. This resulted in a final dataset of 5,942 validated incident records, providing a dependable foundation for all subsequent analysis.

Although the excluded records represented only a small proportion of the available data, validating and documenting their exclusion ensured transparency and strengthened confidence in the integrity of the analytical results. With the database successfully established, Keystra developed an analytical SQL view to serve as the project's reporting layer. Rather than querying the imported table directly, this approach created a consistent and reusable data model from which every SQL investigation and Power BI visualisation would be generated.

Establishing a single analytical layer improved consistency across the project, simplified query development, and ensured that all business insights were derived from the same validated dataset. By the end of this phase, the project had progressed beyond data preparation and into a fully structured analytical environment. The combination of a validated database and a dedicated analytical layer provided the foundation for the next stage of the engagement, where SQL was used to investigate incident patterns, operational risk, asset performance, failure mechanisms, and their business impact. From this point onward, the focus shifted from organising data to generating evidence that could support integrity management, operational planning, and executive decision making.

By this stage of the engagement, Keystra Energy Group had established a validated analytical database capable of supporting a detailed assessment of hazardous liquid pipeline incidents. The focus of the project therefore shifted from preparing data to understanding what the data revealed about pipeline integrity, operational risk, and the factors contributing to incident occurrence.

While the PHMSA dataset provides a comprehensive record of reported incidents, its true value lies in the insights it can generate when viewed through a business lens. Rather than producing isolated statistical outputs, the objective of this phase was to investigate the operational questions that pipeline operators face every day.

Where are incidents occurring most frequently?

Which assets are most vulnerable?

What factors are driving failures?

What are the consequences when incidents occur?

And ultimately, where should integrity management efforts be focused?

To answer these questions, the investigation followed a logical progression. It began by establishing an understanding of the overall incident landscape before examining where operational risk is concentrated, which pipeline assets are most frequently involved, the mechanisms driving failures, and the business consequences that follow. The investigation then assessed operator exposure before bringing these findings together to identify the areas that should be prioritised for integrity management and preventive investment. Each investigation builds upon the previous one, creating a connected narrative rather than a series of independent analyses.

This approach ensures that every finding contributes to a broader understanding of pipeline integrity and provides practical evidence to support operational planning, maintenance prioritisation, and executive decision-making. The sections that follow present the results of this investigation in the same sequence, allowing each stage of the analysis to naturally inform the next.

INVESTIGATION 1: UNDERSTANDING THE INCIDENT LANDSCAPE

Business Question>> What is the overall scale, trend, and distribution of hazardous liquid pipeline incidents across the United States, and what does this reveal about the industry's operational risk profile?

Before identifying where operational risk is concentrated or determining which assets require greater integrity management attention, it is essential to establish a clear understanding of the overall incident landscape. Without this context, individual findings can be difficult to interpret and may not accurately reflect the broader operational environment in which pipeline operators' function. This investigation provides that

foundation by examining the scale, historical trend, and distribution of hazardous liquid pipeline incidents reported to the Pipeline and Hazardous Materials Safety Administration (PHMSA) between 2010 and 2026. Rather than focusing on individual incidents, the objective is to understand the broader characteristics of incident occurrence across the industry, including how frequently incidents occur, how reporting patterns have evolved over time, and how incidents are distributed across operators and geographic regions.

Establishing this baseline is important because it provides the context for every subsequent stage of the investigation. Understanding the overall incident landscape allows later analyses of geographic exposure, asset integrity, failure mechanisms, incident consequences, and operator performance to be interpreted within the wider operational picture. It also enables decision makers to distinguish between isolated observations and recurring patterns that may require strategic intervention.

The sections that follow present a high-level assessment of the hazardous liquid pipeline incident landscape, providing an evidence-based overview of the industry's operational profile before examining the specific drivers of pipeline integrity risk.

1.1 Incident Landscape Overview

The first step in understanding the industry's operational risk profile was to establish the overall scale of hazardous liquid pipeline incidents captured within the PHMSA dataset. This provides an indication of the breadth of the investigation and highlights the extent to which pipeline incidents continue to affect operators across the United States. The below table presents a summary of the dataset used throughout this investigation.

METRIC	VALUE
Total Incidents	5942
Study Period	2010 - 2026
States Represented	49
Pipeline Operators	357
Average Incidents per Year	349.5
Average Incidents per Month	29.1
Average Incidents per Day	0.96

The analytical dataset comprises 5,942 validated hazardous liquid pipeline incidents reported over a seventeen-year period, spanning 2010 to 2026. These incidents involve 357 pipeline operators operating across 49 U.S. states, providing broad industry coverage and a representative view of hazardous liquid pipeline operations within the United States. Although the annual number of reported incidents varies throughout the study period, the dataset indicates an average of approximately 350 incidents each year, equivalent to 29 incidents every month or nearly one reported incident every day.

These figures demonstrate that pipeline failures remain a recurring operational reality rather than isolated occurrences, reinforcing the importance of continuous integrity management and proactive maintenance strategies. Beyond its size, the dataset provides a strong foundation for business analysis because it captures incidents across a wide range of operators, geographic locations, pipeline assets, and operating conditions. This breadth of coverage enables meaningful comparisons throughout the investigation and provides confidence that the patterns identified are representative of broader industry trends rather than individual events.

Establishing this baseline is essential because every subsequent investigation builds upon it. Understanding the overall scale of incident occurrence provides the context needed to evaluate where operational risk is concentrated, which assets experience the greatest exposure, what mechanisms contribute to failures, and the business consequences that follow.

1.2 Incident Trend Analysis

Understanding the overall number of reported incidents provides an important baseline, but it does not explain how the industry's risk profile has evolved over time. Examining long-term incident trends makes it possible to identify whether pipeline integrity performance has improved, deteriorated, or remained relatively stable over the reporting period. It also provides valuable context for assessing the effectiveness of industry-wide maintenance practices, infrastructure investment, and integrity management programmes. The below table summarises the key observations from the annual and monthly incident trends identified during the analysis.

TREND	KEY FINDING
Annual Trend	Incident frequency increased from 333 incidents in 2010 to a peak of 460 incidents in 2015, followed by a gradual decline in subsequent years
Monthly Pattern	Incident activity remained relatively consistent throughout the year, with January (570 incidents) and February (563 incidents) recording the highest number of reported incidents

The annual trend indicates that reported incidents increased steadily during the first half of the study period before reaching their highest level in 2015. From that point onwards, incident frequency generally declined, suggesting a gradual improvement in overall pipeline integrity performance across the industry.

While the PHMSA dataset does not identify the specific reasons for this reduction, the trend may reflect the combined impact of improved inspection programmes, infrastructure replacement, stronger regulatory oversight, advances in leak detection technologies, and more proactive maintenance strategies adopted by pipeline operators.

Despite this encouraging trend, the analysis also shows that hazardous liquid pipeline incidents remain a persistent operational challenge.

Even in years with lower incident counts, hundreds of incidents continued to be reported, indicating that pipeline integrity requires ongoing attention rather than periodic intervention. Monthly incident patterns showed far less variation than annual trends. Although January and February recorded the highest number of reported incidents, incident activity remained relatively consistent throughout the year. This suggests that pipeline failures are not confined to a single operating season but represent a continuous operational risk that requires year-round monitoring, inspection, and maintenance.

Taken together, these findings indicate that while industry performance appears to have improved over time, the operational risk associated with hazardous liquid pipeline systems has not been eliminated. Sustained investment in asset integrity, condition monitoring, and preventive maintenance remains essential to reducing future incident occurrence.

1.3 Industry Distribution

Beyond understanding how incident frequency has changed over time, it is equally important to examine how incidents are distributed across the industry. Identifying where incidents occur, the number of operators involved, and the quality of the reporting data provides additional context for understanding the operational environment before investigating specific sources of risk. Below table summarises the states with the highest reported incident frequencies.

STATES	INCIDENTS	PERCENTAGE OF TOTAL
Texas	2,424	40.79%
Oklahoma	472	7.94%
Louisiana	362	6.09%
California	292	4.91%
Kansas	240	4.04%

The analysis reveals a clear geographic concentration of incidents. Texas alone accounted for more than two-fifths of all reported hazardous liquid pipeline incidents during the study period, significantly exceeding every other state represented in the dataset. Oklahoma, Louisiana, California, and Kansas followed at considerably lower levels but also recorded substantial incident activity.

INVESTIGATION SUMMARY

Before examining where pipeline risks are concentrated or what factors contribute to failures, it was important to establish a clear picture of the industry's overall incident landscape. This investigation showed that hazardous liquid pipeline incidents remain a

recurring operational challenge across the United States, with 5,942 validated incidents reported between 2010 and 2026, involving 357 pipeline operators across 49 states.

Although the number of reported incidents has generally declined since its peak in 2015, the analysis shows that pipeline failures continue to occur regularly, averaging almost 350 incidents each year. This suggests that while improvements in integrity management may have reduced incident frequency over time, maintaining safe and reliable pipeline operations still requires continuous investment, monitoring, and preventive maintenance.

The investigation also highlighted that incident activity is not evenly distributed across the industry. A relatively small number of states account for a significant proportion of reported incidents, reflecting differences in pipeline density, operational activity, and infrastructure exposure. Together, these findings provide the context needed to interpret the more detailed analyses that follow, ensuring that subsequent investigations are viewed within the broader operational picture rather than as isolated observations.

EXECUTIVE RECOMMENDATION

The findings from this investigation reinforce the importance of adopting a risk-based approach to pipeline integrity management. Rather than applying maintenance resources uniformly across an entire network, pipeline operators should use historical incident data to identify areas with greater operational exposure and prioritise inspection, maintenance, and capital investment accordingly. Establishing a clear understanding of the overall incident landscape is only the starting point. The investigations that follow build on this foundation by examining where incidents are most concentrated, which assets are most vulnerable, what factors are driving failures, and the operational consequences that have the greatest impact on pipeline performance.

Together, these insights provide a stronger basis for planning targeted integrity programmes and supporting long term operational resilience. This distribution is consistent with the operational profile of the U.S. energy industry. States with extensive pipeline infrastructure, high production activity, and major transportation corridors naturally experience greater operational exposure, increasing the likelihood of reported incidents. As a result, incident frequency should be interpreted alongside infrastructure density and operational activity rather than as a standalone measure of performance.

The dataset also represents 357 pipeline operators, demonstrating that hazardous liquid pipeline incidents are not confined to a small number of organisations. Instead, the findings highlight that maintaining pipeline integrity is an industry-wide responsibility requiring consistent asset management, inspection programmes, and operational oversight across operators of different sizes and operating environments.

An assessment of the reporting profile further strengthens confidence in the quality of the analytical findings. More than 96% of all incident records were classified as Original

Final or Supplemental Final reports, indicating that the vast majority of incidents had completed the regulatory investigation process before being included in the analysis. This provides greater assurance that reported causes, consequences, and financial estimates reflect validated information rather than preliminary assessments.

Taken together, these findings show that while hazardous liquid pipeline incidents occur across a broad range of operators and geographic locations, operational exposure is not evenly distributed. Certain regions account for a disproportionately large share of reported incidents, highlighting the importance of geographically targeted integrity programmes and risk-based resource allocation rather than uniform maintenance strategies.

INVESTIGATION 2: GEOGRAPHIC RISK ASSESSMENT

Business Question>> Where is pipeline operational risk concentrated geographically, and which regions require greater attention from a pipeline integrity management perspective?

Following the assessment of the overall hazardous liquid pipeline incident landscape, the next stage of the investigation focused on understanding how geographic location influences pipeline risk exposure. The previous investigation established the scale and distribution of incidents across the United States. However, incident frequency alone does not provide a complete understanding of geographic risk. A region may experience a high number of incidents because of extensive pipeline infrastructure and operational activity, while another region with fewer incidents may present greater concern if individual failures result in significant environmental damage, financial losses, or safety consequences.

To support a more informed integrity management strategy, this investigation evaluates geographic exposure beyond incident counts by examining three key dimensions:

- * Environmental exposure —> the volume of hazardous liquid released and not recovered following incidents.
- * Safety exposure —> the human consequences associated with reported incidents, including fatalities and injuries.
- * Geographic risk prioritisation —> the combined assessment of operational frequency and incident consequences.

The objective is to identify regions that require different forms of risk management attention, whether due to frequent operational exposure or the potential severity of individual failures.

2.1 Environmental Exposure by State

Which states present the greatest environmental exposure following hazardous liquid pipeline incidents?

Environmental impact was assessed by evaluating the volume of product released and the effectiveness of recovery efforts. The analysis focused on three measures:

* **Total Unintentional Release (BBLs):** The total volume of hazardous liquid released during incidents.

* **Recovered Barrels (BBLs):** The quantity of released product recovered during response activities.

* **Net Unrecovered Barrels (BBLs):** The remaining product after recovery efforts, representing potential environmental exposure.

STATE	NET UNRECOVERED BARRELS
Texas	365,043.09
Louisiana	65,914.40
Mississippi	51,894.96
West Virginia	44,996.45
Illinois	36,180.53
Oklahoma	31,183.16
North Dakota	26,842.33
Kansas	24,273.75

Texas recorded the highest volume of unrecovered product by a significant margin, with more than 365,000 barrels remaining unrecovered following reported incidents. This reflects both the scale of pipeline operations within the state and the high level of incident activity identified during the previous investigation.

However, the analysis also highlights regions where environmental exposure is driven by incident consequence rather than incident frequency alone. West Virginia, despite recording a relatively small number of incidents, ranked among the highest states for unrecovered product volume. Similarly, Mississippi demonstrated significant environmental exposure despite having considerably fewer incidents than the highest frequency states.

These findings demonstrate that environmental risk is not always proportional to incident frequency. A smaller number of high-consequence events can create substantial environmental impact and require targeted mitigation strategies.

Executive Implication>>

Geographic integrity planning should consider both the frequency and severity of pipeline incidents. Regions with high unrecovered product volumes should receive enhanced spill preparedness, improved containment capability, and stronger environmental monitoring to reduce the long-term impact of hazardous liquid releases.

2.2 Safety Exposure by State

Which geographic regions experience the greatest human safety consequences following pipeline incidents?

Although environmental and financial impacts represent significant areas of concern, pipeline integrity management must also consider the potential consequences to personnel and surrounding communities. Safety exposure was assessed using:

- * Total fatalities
- * Total injuries
- * Combined casualties (fatalities + injuries)
- * Average casualties per incident

STATES	FATALITIES
Texas	8
Alabama	2
Illinois	2
Louisiana	2
Nebraska	1
Georgia	1

Texas also recorded the highest number of injuries, with 34 reported cases during the study period. Compared with environmental and financial impacts, human casualties were relatively limited across the dataset. The majority of states recorded no fatalities or injuries associated with reported hazardous liquid pipeline incidents.

However, the absence of widespread casualties does not indicate the absence of operational risk. Pipeline failures can still result in significant consequences, including environmental damage, financial losses, operational disruption, regulatory exposure, and reputational impact. Texas recorded the highest number of casualties, which aligns with its position as the state with the largest incident volume. However, when considered relative to its number of reported incidents, the average casualty rate remained low, suggesting that most incidents primarily affected infrastructure and the environment rather than resulting in direct human harm.

Executive Implication>>

Pipeline integrity strategies should continue to evaluate safety outcomes alongside environmental and operational consequences. Maintaining low casualty rates remains important, but risk management decisions should not rely solely on human safety indicators because incidents without injuries can still create significant business impacts.

2.3 Geographic Risk Prioritisation

Which geographic regions should receive greater attention based on combined operational exposure and incident consequences?

The previous sections assessed environmental and safety exposure independently. This final stage combines multiple dimensions to provide a broader understanding of geographic risk. The prioritisation framework considers:

- * **Operational Exposure:** Historical incident frequency.
- * **Financial Exposure:** Estimated financial consequences associated with incidents.
- * **Environmental Exposure:** Net unrecovered product volumes.
- * **Safety Exposure:** Fatalities and injuries.

STATE	OPERATIONAL EXPOSURE	FINANCIAL EXPOSURE	ENVIRONMENTAL EXPOSURE	SAFETY EXPOSURE	OVERALL PRIORITY
Texas	Very high	High	Very high	Moderate	Critical
Louisiana	High	High	High	Low	High
Kansas	High	High	High	Low	High
Michigan	Low	Very high	Moderate	Low	High Consequences
West Virginia	Very low	High	Very high	Low	High Consequences
Illinois	Moderate	High	High	Moderate	High
Oklahoma	High	Moderate	High	Low	High
North Dakota	Moderate	High	High	Low	High

The geographic risk assessment demonstrates that pipeline exposure cannot be evaluated using incident frequency alone. Texas represents the most significant operational hotspot due to its combination of high incident frequency, substantial environmental exposure, and recorded safety impacts.

However, Michigan and West Virginia highlight a different type of risk profile, where fewer incidents have resulted in disproportionately severe financial or environmental consequences. This distinction is important because different risk profiles require different management strategies. High frequency regions may require stronger monitoring and preventive maintenance programmes, while high consequence regions may require targeted engineering assessments, improved response planning, and focused mitigation activities.

Executive Implication>>

The geographic analysis shows that pipeline risk cannot be measured by incident frequency alone. While states such as Texas experience a high number of incidents because of their extensive pipeline operations, other states with fewer incidents can still pose significant risk when those incidents result in major financial losses or environmental damage. For Keystra Energy Group, this means that maintenance and integrity planning should consider both how often incidents occur and how severe their consequences are. Taking this broader view allows resources to be directed towards the areas that present the greatest overall business risk rather than simply those with the highest incident counts.

INVESTIGATION SUMMARY

This investigation examined pipeline risk from a geographic perspective by looking beyond where incidents occur and assessing their environmental, financial, and safety impacts. The findings showed that different states face different types of risk. Texas recorded the highest level of operational exposure due to its large number of incidents, while states such as West Virginia and Michigan demonstrated that even a small number of incidents can lead to significant environmental or financial consequences.

Overall, the analysis confirms that effective geographic risk management requires balancing both the frequency of incidents and the severity of their impact. This provides a stronger basis for prioritising inspection, maintenance, and emergency response activities across the pipeline network.

EXECUTIVE RECOMMENDATION

Based on these findings, Keystra Energy Group should adopt a risk-based approach to geographic integrity management. States with frequent incidents should continue to receive regular inspections, preventive maintenance, and continuous monitoring to reduce recurring failures. At the same time, states where incidents have historically resulted in high financial or environmental impacts should receive targeted engineering assessments and strengthened emergency response planning. By allocating resources according to both incident frequency and consequence, Keystra can improve asset reliability, reduce operational risk, and make more informed maintenance and investment decisions across its pipeline network.

INVESTIGATION 3: ASSET AND INFRASTRUCTURE RISK ASSESSMENT

Business Question>> Which pipeline assets and infrastructure characteristics are most frequently associated with hazardous liquid pipeline incidents?

Having identified where hazardous liquid pipeline incidents are geographically concentrated, the next stage of the investigation focused on understanding the infrastructure characteristics associated with these events.

For Keystra Energy Group's Asset Integrity team, understanding location alone is not sufficient for effective risk management. Pipeline networks consist of multiple interconnected assets, including transmission pipelines, pump stations, terminals, storage facilities, and associated equipment. Each component operates under different conditions and may present different integrity challenges.

This investigation therefore examines the physical characteristics of the pipeline network involved in reported incidents. The objective is to identify recurring asset patterns that can support inspection prioritisation, maintenance planning, and long-term infrastructure investment decisions. The analysis focuses on five key areas:

- * Pipeline network classification
- * Infrastructure components involved in incidents
- * Commodity transported
- * Asset age profile
- * Engineering characteristics such as pipe diameter

It is important to note that this assessment identifies where incidents are most frequently observed. It does not determine that a specific asset characteristic directly causes failure, as incident occurrence is influenced by multiple operational, environmental, and maintenance factors.

3.1 Pipeline Network Classification

Are hazardous liquid pipeline incidents more prevalent on interstate or intrastate pipeline systems?

Pipeline classification provides an initial understanding of how incident exposure is distributed across different types of pipeline networks. Interstate pipelines typically transport hazardous liquids across multiple states and form critical links within national energy transportation infrastructure.

PIPELINE CLASSIFICATION	INCIDENTS	PERCENTAGE
Interstate	3,780	63.61%
Intrastate	2,162	36.39%

The analysis shows that nearly two-thirds of reported incidents occurred on interstate pipeline systems. Given the scale, length, and strategic importance of interstate networks, this level of exposure highlights the importance of maintaining strong integrity management practices across these critical assets.

However, the higher incident frequency observed on interstate systems does not necessarily indicate poorer performance. It may also reflect the larger volume of infrastructure, greater operational activity, and increased exposure associated with long-distance transportation networks.

Executive Implication>>

Interstate pipeline systems should remain a key focus area for condition monitoring, inspection planning, and preventive maintenance activities due to their importance within the wider energy transportation network.

3.2 Infrastructure Components Most Frequently Involved

Which pipeline infrastructure components are most commonly associated with reported incidents?

Pipeline systems consist of multiple operational components, and understanding where incidents occur provides insight into areas requiring greater integrity attention.

FACILITY COMPONENTS	INCIDENTS
Onshore Pump/Meter Station Equipment & Piping	1,967
Onshore Pipeline (including valve sites)	1,811
Onshore Terminal/Tank Farm Equipment & Piping	1,489
Breakout Tanks & Storage Vessels	581
Offshore Pipeline	45
Underground Storage Equipment	33
Offshore Platform Equipment	16

The majority of reported incidents occurred within onshore infrastructure, particularly pump and meter station equipment, pipeline segments, and terminal facilities. The results indicate that incidents are not limited to the pipeline itself but occur across multiple stages of the hazardous liquid transportation process. Operational facilities that support movement, storage, and transfer activities also represent important areas of integrity exposure.

Although offshore assets recorded significantly fewer incidents, this should be interpreted within the context of the overall asset population and operating environment rather than as an indication of lower inherent risk.

Executive Implication>>

Asset integrity programmes should adopt a network-wide approach by including transmission pipelines, pump stations, terminals, storage facilities, and associated equipment rather than focusing solely on pipeline segments.

3.3 Commodity Profile

Which transported commodities are most frequently associated with hazardous liquid pipeline incidents?

Understanding commodity exposure is important because different products present different operational, environmental, and safety consequences when released.

COMMODITY	INCIDENTS
Crude Oil	3,052
Refined Petroleum Products	1,885
HVL / Flammable or Toxic Fluids	913
Carbon Dioxide	85
Biofuels	7

Crude oil accounted for more than half of all reported incidents within the dataset, reflecting its significant role within the hazardous liquid pipeline network. Refined petroleum products and highly volatile liquids also contributed substantially to the incident profile, highlighting the broad range of operational risks associated with transporting different hazardous materials.

The distribution of incidents across commodities reinforces that integrity management strategies should consider not only the asset itself but also the characteristics of the material being transported.

Executive Implication>>

Inspection priorities, emergency response planning, and mitigation strategies should be aligned with commodity characteristics, as different products present different environmental, operational, and safety risks.

3.4 Asset Age Profile

Are pipeline incidents concentrated among older infrastructure assets?

Asset age is often considered an important factor in pipeline integrity management due to potential degradation mechanisms such as corrosion, fatigue, and material ageing. This analysis examined incident distribution across asset age groups.

ASSET AGE(YEARS)	INCIDENTS
0 - 9	1,801
10 - 19	432

20 - 29	254
30 - 39	271
40 - 49	388
50 - 59	357
60 - 69	352
70 - 79	138
80 - 89	60
90 - 99	33
100+	4

The highest number of reported incidents occurred among assets aged between 0 and 9 years. This does not necessarily indicate that newer pipelines are less reliable. Instead, the pattern may reflect factors such as recent infrastructure additions, increased operational activity, or the overall age distribution of assets represented within the dataset. Older infrastructure, particularly assets within the 40–69-year range, also contributed significantly to incident occurrence.

This demonstrates that operational risk exists across multiple generations of pipeline assets rather than being limited to ageing infrastructure alone.

Executive Implication>>

Pipeline age should not be considered as a standalone indicator of risk. Asset age should be evaluated alongside inspection history, operating conditions, failure mechanisms, and maintenance records to support effective integrity decisions.

Data Limitation>>

The dataset contains incident records but does not provide the total number of pipelines within each age category. Therefore, incident frequency by age group should not be interpreted as direct evidence that one age category is more failure-prone than another.

3.5 Pipe Diameter Profile

Which pipe diameters appear most frequently within reported incident records?

Pipe diameter provides additional engineering context when evaluating pipeline incidents, although it must be interpreted alongside other operational characteristics. The most frequently reported pipe diameters were:

- * 8-inch
- * 12-inch
- * 16-inch * 6-inch

*** 10-inch**

The results show that incidents occurred across a wide range of pipe diameters, with medium diameter pipelines appearing most frequently within the incident records. However, the findings should not be interpreted as evidence that specific pipe diameters are inherently more susceptible to failure. The dataset does not include the total number of pipelines operating within each diameter category, meaning exposure levels cannot be accurately compared.

Executive Implication>>

Pipe diameter should be considered as one factor within a broader integrity assessment framework. Maintenance prioritisation should also incorporate operating pressure, commodity type, asset age, location, and historical failure patterns.

3.6 Personnel Involved in Incident Identification

Who identified or was associated with the reported pipeline incidents?

Understanding personnel involvement provides insight into operational reporting processes and the quality of workforce-related incident data.

CATEGORY	INCIDENTS
Operator Employee	3,176
Contractor	606
Missing/Not Recorded	2,160

Operator employees accounted for the majority of recorded incident identification or involvement. However, a significant proportion of records contained missing information, limiting the ability to draw strong conclusions regarding workforce-related incident patterns. The level of missing data highlights an opportunity to improve reporting consistency and strengthen operational learning from incident records.

Executive Implication>>

Improving workforce-related data capture would support more effective analysis of human factors, operational practices, and safety performance across pipeline operations.

INVESTIGATION SUMMARY

The asset and infrastructure assessment indicates that pipeline incidents are distributed across multiple areas of the hazardous liquid transportation network rather than being limited to a single asset category.

Key findings include:

* Interstate pipeline systems accounted for the majority of reported incidents, representing approximately two-thirds of all records.

- * Onshore pump/meter station equipment, pipeline segments, and terminal facilities represented the largest sources of incident exposure.
- * Crude oil transportation accounted for the highest proportion of incidents, reflecting its significant role within hazardous liquid pipeline operations.
- * Incident records were distributed across both newer and older assets, demonstrating that asset age alone does not explain pipeline failure occurrence.
- * Pipe diameter patterns should be interpreted carefully because incident frequency does not represent failure probability without knowing the total asset population within each category.

Overall, the findings highlight the importance of evaluating pipeline integrity through a combination of asset characteristics, operating conditions, and historical performance rather than relying on individual attributes.

EXECUTIVE RECOMMENDATION

The analysis supports a more targeted asset integrity strategy where inspection and maintenance activities are prioritised according to asset exposure and operational importance. Keystra Energy Group should therefore ensure integrity programmes extend beyond pipeline segments to include supporting infrastructure such as pump stations, terminals, storage facilities, and associated piping systems.

Asset age, commodity type, pipeline classification, and infrastructure category should be incorporated as supporting indicators within integrity assessments. However, these factors should be evaluated alongside inspection history, operating conditions, and known failure mechanisms to provide a more complete understanding of asset risk. By combining asset level intelligence with geographic and incident consequence analysis, Keystra can improve maintenance prioritisation, optimise capital investment decisions, and strengthen the long-term reliability of its pipeline network.

INVESTIGATION 4: FAILURE MECHANISIM AND LEAK CHARACTERISTICS

Business Question>> What failure mechanisms are driving hazardous liquid pipeline incidents, and how can these insights support more effective pipeline integrity management?

The previous investigations identified where hazardous liquid pipeline incidents occur and the assets most frequently involved. While these findings help Keystra Energy Group understand where operational risk exists, they do not explain what is causing these failures. To strengthen pipeline integrity, it is important to understand the conditions and events that lead to incidents. Identifying the most common failure mechanisms allows maintenance activities to move beyond responding to failures and towards preventing them before they occur. This investigation therefore examines the underlying causes of

hazardous liquid pipeline incidents and the characteristics of the leaks that result from those failures. It also evaluates how different failure mechanisms influence financial loss, environmental impact, and human safety.

The objective is not only to identify the most common causes of pipeline failures but also to understand which failure mechanisms create the greatest business risk. These insights enable Keystra Energy Group to prioritise maintenance activities, improve engineering decisions, and focus integrity investments where they are likely to have the greatest impact.

To achieve this objective, the investigation addresses the following areas:

- * Failure Mechanism Profile → identifying the primary causes of reported pipeline incidents.
- * Leak Characteristics → understanding how pipeline failures typically manifest.
- * Financial Impact by Failure Mechanism – > evaluating the financial consequences associated with different causes.
- * Environmental Impact by Failure Mechanism → assessing the environmental exposure resulting from each failure mechanism.
- * Safety Impact by Failure Mechanism → examining the human safety consequences linked to different types of failures.

Together, these analyses provide a comprehensive understanding of the factors driving pipeline incidents and support a more proactive, risk-based approach to pipeline integrity management.

4.1 Failure Mechanism Profile

What are the primary causes of hazardous liquid pipeline incidents?

Understanding why pipeline failures occur is essential for developing an effective integrity management programme. By identifying the most common failure mechanisms, Keystra Energy Group can focus maintenance activities on the issues that contribute most to operational risk.

FAILURE MECHANISMS	INCIDENTS	PERCENTAGE
Equipment Failure	2,702	45.47%
Corrosion Failure	1,299	21.86%
Incorrect Operation	823	13.85%
Material Failure of Pipe or Weld	399	6.71%
Natural Force Damage	261	4.39%
Excavation Damage	197	3.32%
Other Accident Cause	136	2.29%
Other Outside Force Damage	125	2.10%

The analysis shows that pipeline incidents are largely driven by a small number of recurring failure mechanisms. Equipment failure accounted for almost half of all reported incidents, making it the most significant contributor to pipeline failures across the study period. Corrosion ranked second, reinforcing its continued importance as one of the industry's most persistent asset integrity challenges. Incorrect operation also contributed a considerable number of incidents, highlighting that operational practices and human decision-making remain important factors in pipeline performance.

Together, equipment failure, corrosion failure, and incorrect operation accounted for more than 81% of all reported incidents. This concentration suggests that a relatively small number of recurring issues are responsible for the majority of failures observed across the pipeline network. Rather than treating every incident as a unique event, these findings indicate that significant improvements in pipeline reliability can be achieved by focusing engineering resources on the failure mechanisms that occur most frequently.

4.2 Leak Characteristics

How do pipeline failures typically manifest once an incident occurs?

Understanding the physical characteristics of pipeline leaks provides additional insight into how failures develop and where preventive maintenance efforts should be directed.

LEAK TYPE	INCIDENTS	PERCENTAGE
Pinhole	1,397	23.51%
Seal or Packing Failure	1,341	22.57%
Not Reported/Missing	1,226	20.63%
Connection Failure	760	12.79%
Other	638	10.74%
Crack	580	9.76%

Pinhole leaks and seal or packing failures were the most frequently reported leak characteristics. These types of failures are generally associated with gradual equipment deterioration, corrosion, ageing seals, or component wear rather than sudden structural failure.

One notable finding is that approximately one-fifth of all incident records did not include a reported leak type. While this does not affect the overall incident analysis, it limits the level of engineering insight that can be drawn from historical leak behaviour. Improving the completeness of incident reporting would enable more detailed failure trend analysis and strengthen future integrity investigations.

4.3 Financial Impact by Failure Mechanism

Which failure mechanisms generate the greatest financial losses?

FAILURE MECHANISM	TOTAL FINANCIAL LOSS
Material Failure of Pipe or Weld	\$2.11 Billion
Corrosion Failure	\$716.5 Million
Equipment Failure	\$422.0 Million
Natural Force Damage	\$290.2 Million
Excavation Damage	\$238.8 Million
Incorrect Operation	\$221.5 Million
Other Outside Force Damage	\$182.7 Million

Although equipment failure occurred most frequently, material failure of pipe or weld generated the highest financial losses. This demonstrates that incident frequency alone does not determine business risk. Pipe and weld failures occurred less often but typically resulted in more extensive repairs, longer operational disruptions, and higher remediation costs.

Conversely, equipment failures were more common but generally produced lower financial losses per incident. These findings reinforce the importance of evaluating both the likelihood and the consequence of pipeline failures when prioritising integrity investments.

4.4 Environmental Impact by Failure Mechanism

Which failure mechanisms contribute most to environmental exposure?

FAILURE MECHANISM	NET UNRECOVERED PRODUCT
Material Failure of Pipe or Weld	191,767.06
Excavation Damage	181,932.48
Equipment Failure	125,945.15
Other Outside Force Damage	92,271.83
Corrosion Failure	74,058.92
Natural Force Damage	66,962.56
Incorrect Operation	44,681.11
Other Accident Cause	21,382.61

The analysis shows that material failures involving pipes and welds resulted in the largest volumes of unrecovered product, followed closely by excavation damage. Although excavation damage represented a relatively small proportion of reported incidents, the environmental consequences of those incidents were often severe. This demonstrates that

environmental exposure is influenced more by the scale of individual failures than by how frequently they occur. These findings highlight the importance of combining failure frequency with consequence when evaluating environmental risk.

4.5 Safety Impact by Failure Mechanism

Which failure mechanisms present the greatest risk to human safety?

FAILURE MECHANISM	FATALITIES	INJURIES
Excavation Damage	8	12
Incorrect Operation	1	24
Other Outside Force Damage	5	3
Other Accident Cause	1	4
Equipment Failure	0	2
Material Failure of Pipe or Weld	1	1
Corrosion Failure	0	0
Natural Force Damage	0	0

The safety assessment presents a different perspective from the operational, financial, and environmental analyses. While equipment failure and corrosion accounted for most reported incidents, excavation damage resulted in the highest number of fatalities, and incorrect operation produced the highest number of injuries. This indicates that the most common failure mechanisms are not necessarily those with the greatest human consequences. Activities involving excavation and operational practices appear to present elevated safety risks despite their relatively lower occurrence.

Executive Implication>>

The findings demonstrate that pipeline integrity cannot be strengthened through a single maintenance strategy. Different failure mechanisms create different types of operational risk. While equipment reliability remains the greatest opportunity for reducing incident frequency, high-consequence failures such as pipe and weld defects, excavation damage, and operational errors require targeted engineering controls because of their financial, environmental, and safety impacts.

For Keystra Energy Group, understanding these differences enables maintenance resources to be prioritised according to both the likelihood of failure and its potential consequences.

INVESTIGATION SUMMARY

This investigation examined the engineering and operational factors responsible for hazardous liquid pipeline incidents and assessed the consequences associated with each failure mechanism. The analysis showed that equipment failure, corrosion, and incorrect operation account for the majority of reported incidents. However, the findings also

demonstrate that the most frequent failures are not always the most severe. Material failures involving pipes and welds generated the greatest financial losses and environmental impacts, while excavation damage and incorrect operation were associated with the highest human safety consequences. Overall, the investigation highlights the importance of considering both failure frequency and failure consequence when developing pipeline integrity strategies.

EXECUTIVE RECOMMENDATION

Keystra Energy Group should adopt a targeted integrity management approach that addresses each failure mechanism according to its level of operational risk. Preventive maintenance programmes should prioritise reducing equipment failures through enhanced inspection, condition monitoring, and reliability improvement initiatives.

Corrosion management should remain a core element of long-term asset integrity planning, while high-consequence failures involving pipe and weld defects should receive advanced engineering assessments, non-destructive testing, and asset renewal where appropriate. In addition, reducing excavation damage and incorrect operation will require stronger contractor management, improved permit-to-work processes, workforce competency development, and greater operational discipline. Aligning engineering resources with both failure frequency and consequence will strengthen asset reliability, reduce business risk, and support safer, more resilient pipeline operations.

INVESTIGATION 5: CONSEQUENCE ASSESSMENT AND BUSINESS IMPACT

Business Question>> What operational, financial, environmental, and human consequences do hazardous liquid pipeline incidents have on business performance, and what do these impacts mean for pipeline integrity management?

The previous investigation examined the engineering and operational factors responsible for hazardous liquid pipeline incidents, providing insight into why failures occur across the pipeline network. While understanding the root causes of incidents is essential for preventing future failures, effective integrity management also requires understanding the consequences when those failures occur.

For Keystra Energy Group, the impact of a pipeline incident extends far beyond the initial asset failure. Every incident has the potential to disrupt operations, generate significant financial costs, damage the environment, attract regulatory attention, and affect the safety of employees and surrounding communities. The true business impact of an incident is therefore measured not only by how often failures occur, but also by the scale of their consequences. Assessing these consequences enables the organisation to understand where the greatest business risks exist and where investments in prevention, emergency preparedness, and operational resilience are likely to deliver the greatest

value. It also supports more informed decision making by demonstrating that reducing the impact of pipeline failures is just as important as reducing their frequency.

This investigation evaluates the consequences of hazardous liquid pipeline incidents across four key areas:

- * **Financial Impact Assessment** → examining the direct costs associated with reported incidents.
- * **Environmental Consequences** → evaluating product recovery performance and the potential environmental exposure resulting from released hazardous liquids.
- * **Human Impact Assessment** → assessing the effect of pipeline incidents on workforce and public safety.
- * **Incident Severity Profile** → understanding how incidents are distributed based on their financial impact.

Together, these analyses provide a broader understanding of how pipeline failures affect business performance and reinforce the importance of integrating consequence management into Keystra Energy Group's long-term pipeline integrity strategy.

5.1 Financial Impact Assessment

What is the overall financial burden associated with hazardous liquid pipeline incidents?

While pipeline failures are often discussed in terms of incident frequency, their financial consequences provide a clearer picture of the impact on business performance. Beyond repairing damaged infrastructure, organisations must also manage emergency response activities, environmental restoration, product losses, regulatory obligations, and other operational costs. Evaluating these costs helps identify where pipeline incidents place the greatest financial pressure on the business.

COST CATEGORY	TOTAL COST
Property Damage	\$672.12 Million
Environmental Remediation	\$1.67 Billion
Billion Emergency Response	\$1.77 Billion
Product Loss	\$61.33 Million
Other Costs	\$137.60 Million
Total Estimated Incident Cost	\$4.31 Billion

The financial assessment reveals that hazardous liquid pipeline incidents generated an estimated \$4.31 billion in costs during the study period, highlighting the significant economic burden associated with pipeline failures. Emergency response activities represented the largest cost category, accounting for approximately \$1.77 billion, closely followed by \$1.67 billion spent on environmental remediation. These findings indicate that

the greatest financial impact of pipeline incidents occurs after the initial failure, as organisations work to contain spills, restore affected areas, repair damaged infrastructure, and resume normal operations.

By comparison, direct product losses accounted for only \$61.33 million, representing a relatively small proportion of the total financial burden. This demonstrates that the value of the lost product is often outweighed by the broader operational and environmental costs that follow a pipeline incident. From a business perspective, these findings reinforce that pipeline integrity is not solely an engineering concern but also a significant financial consideration. Every avoided incident represents not only improved operational reliability but also the potential to reduce emergency response expenditure, environmental liabilities, and overall business disruption.

5.2 Environmental Consequences

How effective are recovery efforts following hazardous liquid pipeline incidents, and what do the remaining unrecovered releases indicate about environmental exposure?

While preventing pipeline failures remains the primary objective of integrity management, the effectiveness of response and recovery efforts plays an equally important role in reducing the long-term environmental impact of an incident. Measuring the volume of product released alongside the amount successfully recovered provides valuable insight into the scale of environmental exposure and the effectiveness of spill response operations.

ENVIRONMENTAL METRIC	VALUE
Total Product Released	1,342,931.35 BBLs
Total Product Recovered	543,928.63 BBLs
Net Unrecovered Product	799,002.72 BBLs

The analysis shows that more than 1.34 million barrels of hazardous liquid were released during the study period. Of this volume, approximately 543,929 barrels were successfully recovered, leaving an estimated 799,003 barrels unrecovered. Although recovery operations helped contain a substantial proportion of the released product, the volume that remained unrecovered represents a significant environmental concern. Unrecovered releases have the potential to affect soil, water resources, ecosystems, and surrounding communities while increasing remediation costs and regulatory obligations.

For Keystra Energy Group, these findings highlight that the success of pipeline integrity programmes should not be measured solely by their ability to prevent failures. The organisation's ability to respond quickly, contain releases effectively, and minimise environmental damage is equally important in reducing the overall impact of pipeline incidents. Strengthening emergency response capabilities, improving leak detection

systems, and enhancing spill containment procedures can significantly reduce environmental exposure and improve operational resilience when incidents occur.

5.3 Human Impact Assessment

What was the overall impact of hazardous liquid pipeline incidents on human safety?

Protecting people remains one of the most important objectives of pipeline integrity management. Although hazardous liquid pipeline incidents often receive attention because of their operational and environmental consequences, their impact on employees, contractors, emergency responders, and the public is equally important. Assessing fatalities and injuries provides insight into the human cost of pipeline failures and helps determine whether current safety measures are effectively reducing harm.

SAFETY METRIC	VALUE
Total Fatalities	16
Total Injuries	46
Total Casualties	62
Average Casualties per Incidents	0.0104

Over the seventeen-year study period, hazardous liquid pipeline incidents resulted in 16 fatalities and 46 injuries, giving a total of 62 recorded casualties. Relative to the 5,942 incidents analysed, the average casualty rate remained low at 0.0104 casualties per incident. While these figures suggest that most incidents did not result in direct harm to people, every fatality and injury represents a significant human consequence that extends beyond operational and financial performance. Behind each statistic are individuals, families, and communities affected by pipeline failures.

The findings also reinforce an important point observed throughout this investigation: the severity of an incident cannot be measured by financial loss or environmental impact alone. Some incidents may result in limited economic damage yet have serious safety consequences, while others may generate substantial financial costs without causing injuries. For Keystra Energy Group, maintaining a low casualty rate is encouraging, but it should not reduce the emphasis placed on safety. Continuous investment in operational discipline, workforce competency, emergency preparedness, and robust safety management systems remains essential to protecting people while strengthening the overall resilience of pipeline operations.

5.4 Incident Severity Profile

How are hazardous liquid pipeline incidents distributed based on their financial impact, and what does this reveal about overall business risk? Not every pipeline incident has the same level of impact. While many incidents can be contained before they escalate, a smaller number result in substantial financial losses and operational disruption. Understanding how incidents are distributed across different severity levels provides a

clearer picture of the organisation's overall risk exposure and helps identify where integrity management efforts can deliver the greatest value.

SEVERITY LEVEL	INCIDENTS	PERCENTAGE
Low Impact	4,186	70.45%
Medium Impact	1,356	22.82%
High Impact	400	6.73%

The analysis shows that the majority of hazardous liquid pipeline incidents were classified as low-impact events, accounting for over 70% of all reported incidents. This suggests that many failures were managed before developing into major operational or financial events. However, the remaining incidents present a different picture. Nearly 23% were classified as medium-impact events, while 400 incidents (6.73%) resulted in high financial impact. Although these high-impact incidents occurred relatively infrequently, they have the potential to account for a disproportionate share of overall business losses.

These findings reinforce an important principle of pipeline integrity management: reducing incident frequency alone is not enough. Organisations must also focus on preventing the relatively small number of high-consequence events that can lead to extensive financial loss, environmental damage, prolonged operational disruption, and increased regulatory scrutiny.

For Keystra Energy Group, this highlights the importance of adopting a risk-based approach that prioritises assets and operating conditions with the greatest potential to produce severe outcomes, rather than treating every incident as carrying the same level of risk.

Executive Implication>>

The consequence assessment demonstrates that the true impact of pipeline incidents extends far beyond the initial equipment failure. Financial losses, environmental exposure, operational disruption, and human safety outcomes each contribute to the overall business risk faced by pipeline operators. For Keystra Energy Group, this means that measuring success should not be based solely on reducing the number of incidents. Equal emphasis should be placed on limiting the severity of incidents when they do occur. Integrating prevention, preparedness, and rapid response into pipeline integrity programmes will enable the organisation to reduce both the likelihood and the consequences of future failures.

INVESTIGATION SUMMARY

This investigation evaluated the business consequences of hazardous liquid pipeline incidents by examining their financial, environmental, and human impacts, together with the overall severity of reported events. The analysis estimated a total financial burden of

\$4.31 billion, with emergency response and environmental remediation representing the largest cost categories. It also revealed that almost 800,000 barrels of hazardous liquid remained unrecovered, highlighting the significant environmental challenges associated with major release events.

Although casualties were relatively low across the study period, the findings reinforce that protecting people remains a fundamental objective of pipeline integrity management. Finally, the severity assessment showed that while most incidents resulted in relatively low financial impact, a small proportion of high-impact events accounted for the greatest business risk. These findings demonstrate that effective integrity management must focus not only on reducing incident frequency but also on preventing the severe events that have the greatest operational, financial, and environmental consequences.

EXECUTIVE RECOMMENDATION

Based on these findings, Keystra Energy Group should continue strengthening its pipeline integrity programme through a balanced approach that combines incident prevention with consequence management. Preventive maintenance and asset integrity initiatives should be supported by enhanced emergency response capabilities, improved leak detection systems, stronger environmental containment measures, and continuous workforce safety programmes.

Particular attention should be given to identifying and mitigating conditions that could lead to high-impact incidents, as these events represent the greatest threat to business performance and operational resilience. By combining proactive maintenance with effective response and recovery strategies, Keystra Energy Group can reduce financial losses, minimise environmental impacts, protect human life, and build a more resilient pipeline network capable of supporting safe and reliable operations over the long term.

INVESTIGATION 6: STRATEGIC RISK INTELLIGENCE AND DECISION FRAMEWORK

The previous investigations examined hazardous liquid pipeline incidents from multiple perspectives, providing insight into where incidents occur, which assets are most frequently affected, the factors driving failures, and the financial, environmental, and safety consequences that follow. Having established these findings, the next step is to translate them into business decisions. The purpose of this investigation is to consolidate the evidence gathered throughout the analysis into a practical decision framework that can support pipeline integrity planning, maintenance prioritisation, and long-term investment across Keystra Energy Group's pipeline network.

Rather than introducing new analysis, this chapter brings together the key insights from the project and identifies the strategic priorities that should guide future operational and asset integrity initiatives.

6.1 Executive Risk Assessment

Based on the findings from this investigation, where should Keystra Energy Group focus its pipeline integrity efforts?

The investigation identified several consistent patterns that require strategic attention. Operational risk is concentrated within a relatively small number of geographic regions, with Texas representing the highest level of incident activity. Interstate transmission pipelines and major onshore facilities, including pump stations and terminal infrastructure, were the assets most frequently associated with reported incidents.

The analysis also showed that equipment failure and corrosion remain the primary drivers of pipeline incidents, while failures involving pipes and welds, excavation activities, and operational errors often resulted in the greatest financial, environmental, or safety consequences. Taken together, these findings demonstrate that effective integrity management requires more than reducing incident frequency.

Maintenance decisions should also consider the potential consequences of failure, allowing resources to be directed towards assets and operating conditions that present the greatest overall business risk.

6.2 Strategic Business Priorities

The findings from this project support five strategic priorities for Keystra Energy Group. Priority 1 —> Adopt a Risk-Based Geographic Inspection Strategy Inspection programmes should focus on operating regions with sustained incident activity and elevated environmental exposure. Prioritising inspections according to risk enables maintenance resources to be allocated where they are likely to deliver the greatest operational value.

Priority 2 —> Improve the Reliability of Critical Pipeline Assets Interstate transmission pipelines, pump stations, and terminal facilities should remain the primary focus of asset integrity programmes. Combining routine inspections with condition monitoring and targeted asset renewal will improve reliability across critical infrastructure.

Priority 3 —> Strengthen Preventive Maintenance Programmes The high proportion of incidents attributed to equipment failure and corrosion highlights the need for stronger preventive maintenance. Expanding equipment inspections, corrosion monitoring, and condition-based maintenance will help identify deterioration before failures occur.

Priority 4 —> Reduce High-Consequence Operational Risks Although less frequent, failures involving pipes, welds, excavation activities, and operational errors generated the most significant business impacts. These risks should be addressed through enhanced engineering assessments, improved contractor management, strengthened operational controls, and periodic competency reviews.

Priority 5 —> Enhance Emergency Response Capability The volume of unrecovered hazardous liquid recorded throughout the study demonstrates the importance of rapid and effective incident response. Continued investment in leak detection, spill

containment, emergency planning, and response readiness will help minimise environmental impacts and improve organisational resilience.

6.3 Strategic Decision Framework

STRATEGIC FOCUS AREA	STRATEGIC OBJECTIVE
Geographic Risk Management	Prioritise inspections with high-risk operating regions
Critical Asset Management	Improve the reliability of transmission pipelines, pump stations, and terminal facilities
Preventive Maintenance	Reduce recurring equipment and corrosion related failures through proactive maintenance
High Consequence Risk Management	Minimize financial, environmental, and safety impacts associated with severe pipeline failures
Emergency Preparedness	Strengthen spill response capability and improve performance following pipeline incidents

EXECUTIVE RECOMMENDATION

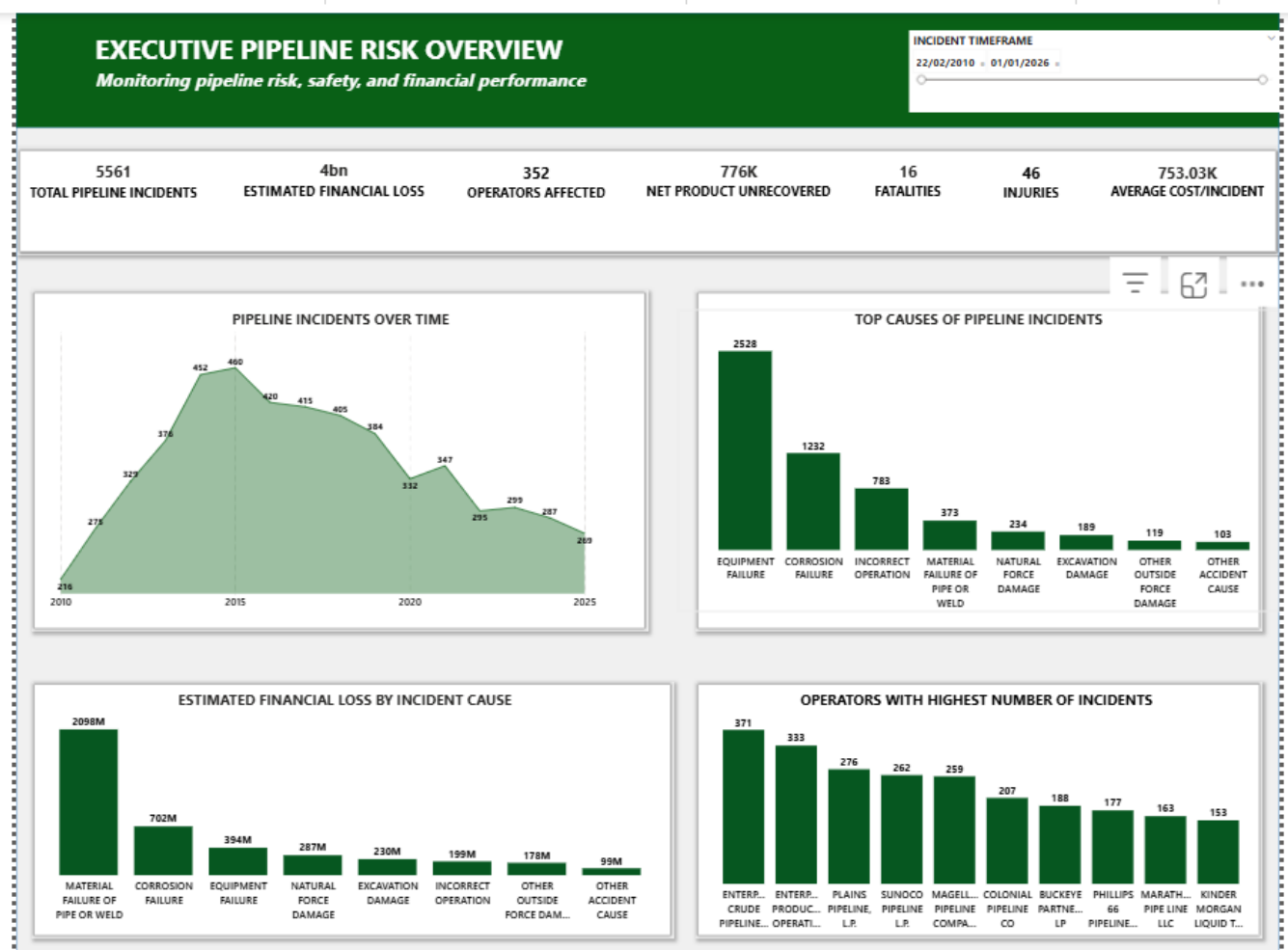
The analysis demonstrates that pipeline integrity is most effective when maintenance decisions are guided by operational evidence rather than routine schedules alone. For Keystra Energy Group, this means directing investment towards the locations, assets, and failure mechanisms that contribute most to overall business risk while strengthening preventive maintenance and emergency response capabilities. Implementing these priorities will support more effective resource allocation, improve asset reliability, reduce operational disruption, and strengthen environmental stewardship across the pipeline network.

The strategic priorities presented in this investigation provide the foundation for the executive dashboard developed in the next chapter, where these insights are transformed into interactive performance indicators that support ongoing operational monitoring and executive decision making.

The six investigations carried out throughout this project answered Keystra Energy Group's most important business questions. They revealed where pipeline incidents occur most frequently, the infrastructure most affected, the dominant causes of failure, and the financial, environmental, and safety consequences associated with those failures. While these findings provide valuable insight, executives do not make decisions by reading lengthy analytical reports every day. They require timely access to the most important information in a format that supports rapid decision-making.

To bridge the gap between detailed analysis and day-to-day operational management, an interactive executive dashboard was developed using Microsoft Power BI. The dashboard transforms the analytical findings into a practical decision-support tool that enables leadership to monitor pipeline integrity, evaluate operational performance, identify emerging risks, and prioritise maintenance activities from a single reporting environment. Rather than introducing new findings, the dashboard brings together the key insights generated throughout this investigation into an interactive platform designed to support continuous monitoring and evidence-based decision-making.

EXECUTIVE DASHBOARD OVERVIEW



The dashboard was developed as the final deliverable of this project and consists of two complementary reporting pages. The first page provides executives with a high-level overview of pipeline performance, bringing together the most critical operational, financial, environmental, and safety indicators identified during the investigation. The second page focuses on pipeline integrity and operational intelligence by enabling engineering and asset management teams to explore infrastructure characteristics, pipeline assets, geographic distribution, and operational performance in greater detail.

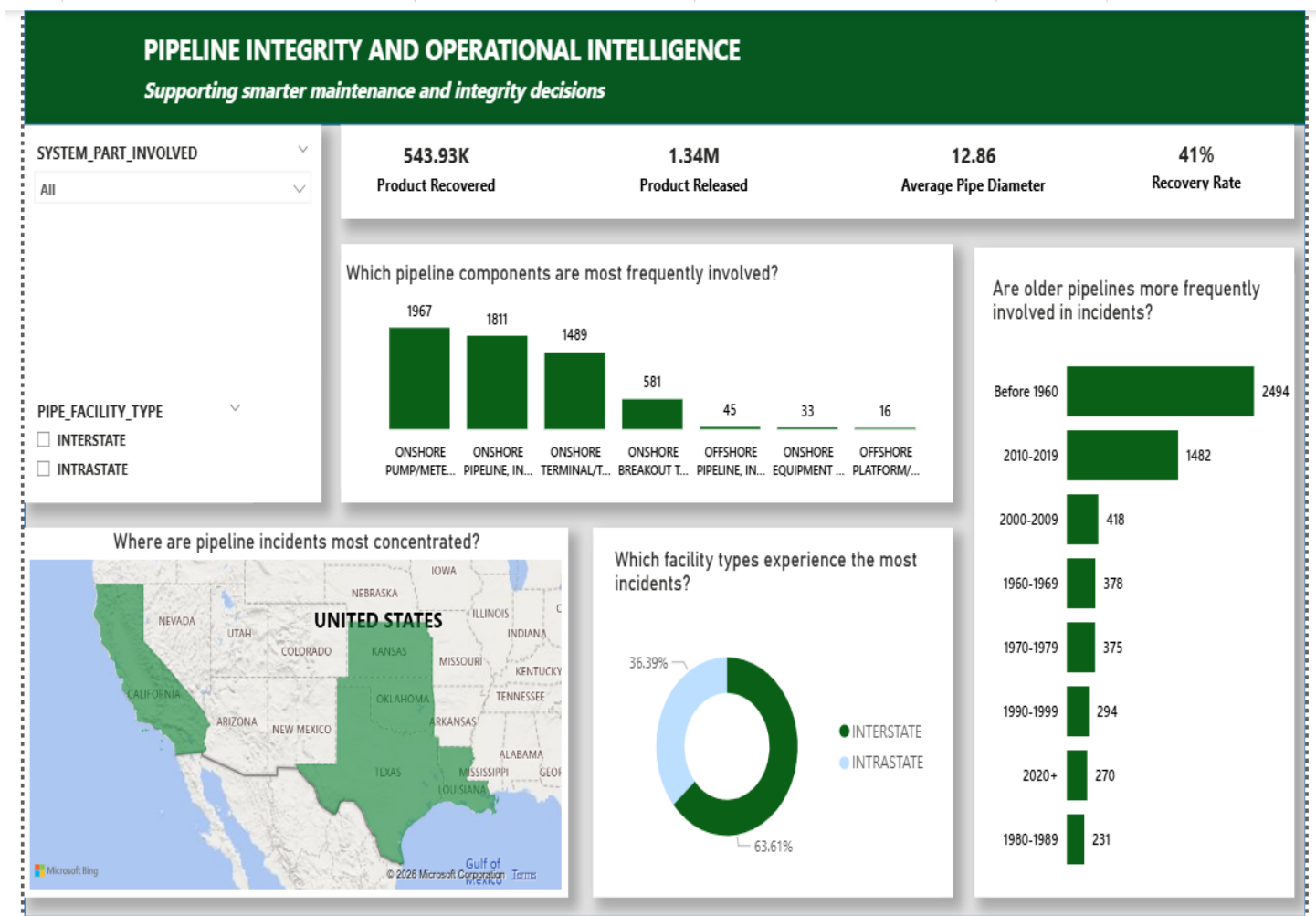
Together, both dashboard pages provide a balanced view of strategic performance and operational risk, allowing users to move from high-level business monitoring to more detailed operational analysis without leaving the reporting environment.

The Executive Pipeline Risk Overview dashboard was designed for senior management and business leaders responsible for monitoring organisational performance. The dashboard consolidates the project's most important performance indicators into a single view, enabling executives to understand the overall state of the pipeline network at a glance. Key metrics relating to incident frequency, financial exposure, environmental impact, operator performance, and workforce safety are presented alongside interactive visualisations that highlight long-term incident trends, leading failure mechanisms, financial losses by incident cause, and operators with the highest number of reported incidents. By bringing these indicators together in one location, the dashboard allows management to quickly identify areas requiring attention while monitoring overall pipeline integrity performance over time.

PIPELINE INTEGRITY AND OPERATIONAL INTELLIGENCE

While the first dashboard focuses on executive oversight, the Pipeline Integrity and Operational Intelligence dashboard supports operational decision-making. The dashboard provides deeper insight into the characteristics of the pipeline network by presenting information on infrastructure components, pipeline classifications, asset age, geographic distribution, and product recovery performance. Interactive filters allow users to examine incident patterns across different facility types and system components, supporting more targeted investigation and maintenance planning. This page enables integrity engineers and maintenance planners to identify infrastructure requiring greater attention while supporting more informed inspection scheduling and asset management decisions. The dashboard represents the practical outcome of the analytical investigation undertaken throughout this project. Instead of relying on static reports, Keystra Energy Group can use this interactive platform to continuously monitor pipeline integrity, evaluate operational performance, identify recurring risks, and support proactive maintenance planning. By integrating executive performance indicators with operational intelligence, the dashboard enables faster decision making, improves visibility across the pipeline network, and supports a more risk informed approach to asset integrity management. Ultimately, it demonstrates how business intelligence can transform historical operational data into

actionable insights that improve reliability, strengthen resilience, and support long-term organisational performance.



This solution has provided Keystra Energy Group with a deeper understanding of pipeline integrity by examining historical incident data, identifying the assets most affected, uncovering the dominant causes of failure, and evaluating their operational, financial, environmental, and safety impacts.